What is PowerShell?

PowerShell is a task automation and configuration management framework from Microsoft, consisting of a command-line shell and a scripting language. It is built on the .NET framework (now .NET Core) and designed especially for system administrators and power users to automate tasks and manage systems efficiently.

Key Features of PowerShell

- **Cmdlets**: Lightweight commands used in the PowerShell environment, typically in Verb-Noun format (e.g., Get-Process, Set-Service).
- **Scripting Language**: PowerShell scripts (.ps1 files) can automate complex workflows, combine multiple cmdlets, control flow with loops and conditionals, and handle errors.
- **Object-Oriented**: Unlike traditional shells that deal with text output, PowerShell works with .NET objects, enabling structured and rich data manipulation.
- **Pipeline**: Cmdlets output objects, which can be piped (|) as input to other cmdlets, allowing powerful data processing chains.
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Remoting: PowerShell supports remote execution of commands across multiple machines via PowerShell Remoting using WS-Management protocol.

- **Modules**: Extensible with modules that add cmdlets and functions, such as Active Directory, Exchange, Azure, and more.

- **Integrated Scripting Environment (ISE)**: GUI for writing, testing, and debugging PowerShell scripts.

- **Cross-platform**: PowerShell Core (PowerShell 7+) runs on Windows, Linux, and macOS.

Basic PowerShell Concepts

- **Cmdlet syntax**:
  Verb-Noun -ParameterName ParameterValue
  Example: Get-Service -Name "wuauserv"

- **Variables**: Denoted with $ sign, e.g., $userName = "Alice"

- **Data types**: PowerShell supports strings, integers, arrays, hash tables (dictionaries), objects, etc.
- **Pipeline**:
  Example: Get-Process | Where-Object {$_.CPU -gt 100}
  This gets processes consuming CPU > 100.
- **Conditional statements**: if, else, switch
- **Loops**: for, foreach, while, do-while

PowerShell Remoting

Allows running commands on remote systems. Commands include:

- Invoke-Command – run command/script block remotely
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Enter-PSSession – start interactive remote session

- Uses WS-Management protocol (WinRM)

Modules and Snap-ins

Modules package cmdlets and functions. You can list modules with:

powershell

Copy code

Get-Module -ListAvailable Import-Module ModuleName

Error Handling

- Try, Catch, Finally blocks allow robust error handling in scripts.
- -ErrorAction parameter controls error behavior (Continue, Stop, SilentlyContinue).

Common Cmdlets

- Get-Help: Show help info about cmdlets
- Get-Command: List cmdlets and functions
- Get-Service: View system services
- Get-Process: List running processes
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- Set-ExecutionPolicy: Set script execution policies
- New-Item, Remove-Item: File system operations
- Get-EventLog: View Windows event logs

Execution Policy

PowerShell restricts script execution by default for security:

- Restricted (default): no scripts run
- RemoteSigned: scripts from internet must be signed
- Unrestricted: all scripts allowed

Advanced Features

- **Desired State Configuration (DSC)**: Declarative configuration management framework.
- **PowerShell Jobs**: Run background tasks asynchronously.
- **Custom functions and scripts**: Extend functionality.
- **Interoperability**: Can call native commands, .NET classes, and COM objects.